

Series and infinite products – 6

Series in number theory: digits, primes, irrationality

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Theoretical minimum.

Theorem 1 (Euler product). For $s > 1$

$$\sum_{n \geq 1} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1},$$

the identity being nothing but unique factorisation written as an expansion of geometric progressions. Letting $s \rightarrow 1^+$ gives both the infinitude of the primes and the divergence of $\sum \frac{1}{p}$, together with the rate $\sum_{p \leq x} \frac{1}{p} = \ln \ln x + O(1)$.

Theorem 2 (Kempner). If from the harmonic series one deletes all n whose decimal expansion contains a fixed digit, the remaining series converges: among the numbers with k digits at most $9 \cdot 8^{k-1}$ survive, and each of them is at least 10^{k-1} , so the block sums are dominated by a geometric progression with ratio $\frac{8}{10}$.

Theorem 3 (irrationality by approximation). A real x is irrational if and only if there are integers p_n, q_n with $0 < |q_n x - p_n| \rightarrow 0$. For a series this means: multiply by a suitable factorial or a suitable power, and show that the tail is strictly between 0 and 1.

Fact 1 (digits count geometrically). Any restriction on the digits of n leaves c^k numbers with k digits for some $c < 10$, and a series $\sum f(n)$ over such n is compared block by block with $\sum (c/10)^k$ -type progressions. The same counting explains why a weight like $a^{(\text{number of zeros in } n)}$ changes the answer only through the value of a .

Fact 2 (Dirichlet series as bookkeeping). Multiplying $\sum \frac{f(n)}{n^s}$ and $\sum \frac{g(n)}{n^s}$ gives the Dirichlet convolution $\sum \frac{(f * g)(n)}{n^s}$; hence $\sum \frac{d(n)}{n^s} = \zeta(s)^2$, $\sum \frac{\mu(n)}{n^s} =$

$\frac{1}{\zeta(s)}$ and $\sum \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}$. All the interchanges are those of Series and infinite products – 5.

1. Prove the Euler product for $\zeta(s)$, $s > 1$, and deduce from it that $\sum_p \frac{1}{p}$ diverges. Where exactly does the argument use unique factorisation?
2. Prove Kempner's theorem for the digit 9 and give an upper bound for the sum of the resulting series; then explain why deleting the numbers containing the block 2026 also leaves a convergent series.
3. (VJIMC 1997) Find all real numbers $a > 0$ for which the series $\sum_{n=1}^{\infty} \frac{a^{f(n)}}{n^2}$ is convergent, where $f(n)$ denotes the number of zeros in the decimal expansion of n .
4. (MΦTH 2002) Let $(\varepsilon_n)_{n \geq 0}$ be a sequence with $\varepsilon_n \in \{-1, 1\}$ for all n . Can the number $\sum_{n=0}^{\infty} \frac{\varepsilon_n}{n!}$ be rational?
5. (Putnam 2017) Suppose that $f(x) = \sum_{i=0}^{\infty} c_i x^i$ is a power series for which each coefficient c_i is 0 or 1. Show that if $f(2/3) = 3/2$, then $f(1/2)$ must be irrational.